

JAYPEE UNIVERSITY OF ENGINEERING & TECHNOLOGY, GUNA
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Course: Computer Network Lab

Course Code: 18B17CI471

B. Tech. (CSE IV Sem.)

Experiment # 2

Aim: Use of crossover and Straight Ethernet cable

1. **Connect Two Computers using a Crossover Cable:** Ever needed to connect two computers together to transfer large amount of data. Use a crossover cable to connect two computers and transfer data.

In order for this setup to work, there are a few things you need to make sure are setup or configured properly.

Step 1 – Configure IP Addresses

Usually, if you are using a crossover cable to connect two computers, the computers are not connected to a LAN network. In this case, you will need to configure static IP addresses for each computer.

You have to make sure that both computers have IP addresses that are on the same subnet. For example, if you give one computer an IP address of 192.168.0.1, then you should give the second computer an IP of 192.168.0.2.

If you used the above IP addresses, you should use a subnet mask of 255.255.255.0. If you use an incorrect subnet mask, the connection will not work.

Lastly, the value for the default gateway should be the same on both machines. You can pick which IP address you want to use, but use it on both computers for the default gateway.

Below is how my setup looks for one computer:

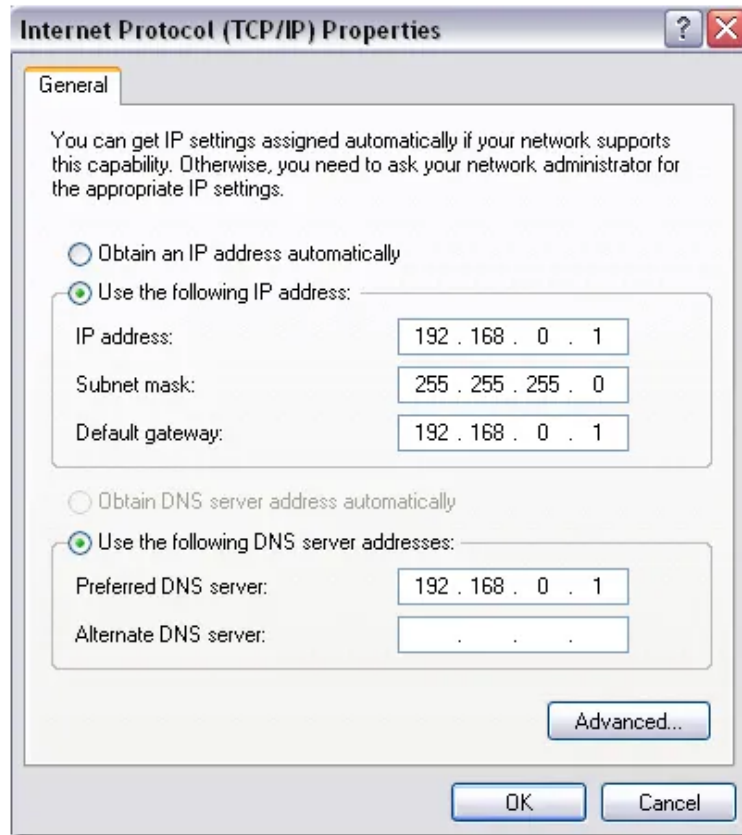


Figure-1

It's worth noting that you could technically leave the Default Gateway and Preferred DNS server fields blank as you really don't have to connect to any device outside the local network.

Step 2 – Crossover Cable

The second thing you need to verify is that you actually have a proper crossover cable. Basically, there are a few different types of crossover cables, but it's best to get the standard crossover cable where the green and orange pairs are swapped and the brown and blue pairs stay in the same position.

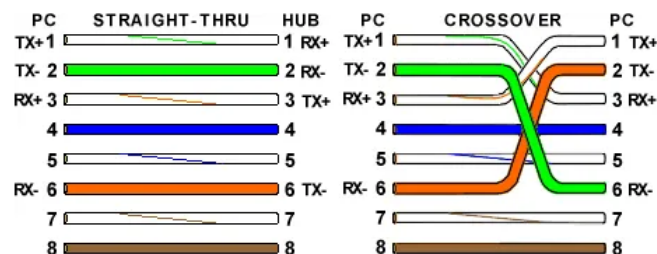


Figure-2

The only pins that need to be crossed are 1, 3 and 2, 6. So just check your cable on both end and make sure it matches the diagram.

Note:- You can use 10 ft crossover cable or you can get **crossover adapters**, which can turn any Ethernet cable into a crossover cable.

Step 3 – Local User Accounts

If you're having problems accessing files both ways, you may have to create a user account on each computer that has the same name and same password. This helps get past any kind of Access is Denied errors that may pop up.

Make sure the accounts are administrator accounts. It is best to create a new account rather than rename a current account.

Step 4 – Disable Firewalls

Finally, you should disable the firewall on both computers to ensure that none of the file sharing ports is being blocked. You can do this by going to Control Panel and then Windows Firewall. Click on Off.

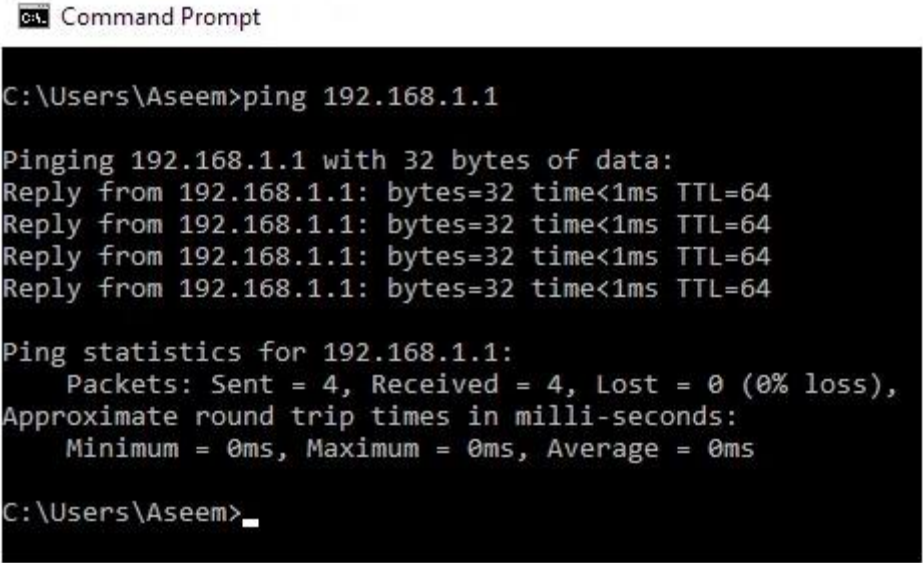


Figure-3

If you follow all these steps, each computer should be able to access the other properly.

Check the connectivity:-

Check the connection by opening a command prompt (click on Start and type in cmd) and trying to ping the other computer. Just type the word ping followed by the IP address of the other computer.



```
Command Prompt

C:\Users\Aseem>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Aseem>
```

Figure-4

You should now be able to go to My Network Places or Network in Windows Explorer and access the other computer. You'll also need to setup a shared folder if you want to transfer files.